



**SPARTAN
INTERNATIONAL
SR. SEC. SCHOOL**

Computer Science Investigatory Project

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CLASS: 12th 'A'

SUBJECT: Computer Science

TOPIC: Mini Super Market Application

Signature of the Subject teacher

Signature of the Student

Acknowledgement

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Lastly, I extend my gratitude to everyone who directly or indirectly contributed to the successful completion of this investigatory project.

Certificate

This is to certify that **Shaafiq N** has successfully completed the Computer Science Investigatory Project on the topic “**Mini Super Market Application**” during the academic year **2026–27** under the guidance of the English teacher.

This project is a genuine piece of work carried out by the student as part of the academic requirements. The project reflects sincere effort, research, and understanding of the topic.

Teacher's Signature: _____

Student's Signature: _____

Date: _____

Index

❖ INTRODUCTION.....	5
❖ DATABASE FOR THE SUPER MARKET APPLICATION.....	6
❖ PROGRAM OR CODE FOR THE SUPER MARKET APPLICATION.....	8
❖ OUTPUT.....	23
❖ BIBLIOGRAPHY	29
❖ CONCLUSION	30

❖INTRODUCTION

The Super Market Management System is a Python-based application developed to simplify product management in a supermarket. It is connected to a MySQL database that stores and manages product information efficiently. The system provides functions such as adding products, removing products, updating product details, searching for products, checking expired products, viewing available stock, and generating customer bills.

The primary objective and basic agenda of this project is to ensure customer safety by clearly displaying in the generated bill whether a product is expired, not expired, or has no expiry date. This helps customers and shopkeepers verify that the product being sold is safe for consumption or use.

The application automatically checks product expiry dates and displays their status during billing. This helps prevent the sale of expired products and improves inventory management.

The project has been developed using Python and MySQL. Although the current version uses a text-based interface, it can also be implemented as a Graphical User Interface (GUI) with tools such as Tkinter, PyQt, or CustomTkinter. A GUI version would provide a more user-friendly and professional experience while maintaining the same functionality.

Overall, this project demonstrates the practical application of programming and database management in supermarket operations.

❖ DATABASE FOR THE SUPER MARKET APPLICATION

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sakila |
| super_market |
| sys |
| world |
+-----+
7 rows in set (0.03 sec)

mysql> show tables from Super_Market;
+-----+
| Tables_in_super_market |
+-----+
| products |
+-----+
1 row in set (0.02 sec)

mysql> use Super_Market;
Database changed
mysql> select * from Products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	NULL
12	Glue	5	10.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

```
14 rows in set (0.00 sec)
```

The above picture displays the MySQL database used in the project. It shows the list of available databases, including the **Super_Market** database created specifically for this application. The picture also shows the **Products** table stored within the database. The table contains details of all products available in the supermarket, including Product ID, Product Name, Total Quantity Available, Individual Product Price, and Expiry Date. This data is used by the Python program to perform operations such as billing, product searching, stock management, expiry checking, and product updates. The records

displayed confirm that the database has been successfully populated with sample product information for testing and implementation purposes.

```
mysql> describe products;
```

Field	Type	Null	Key	Default	Extra
Product_ID	bigint unsigned	NO	PRI	NULL	auto_increment
Product_Name	char(50)	NO	PRI	NULL	
Total_Quantity_Available	bigint unsigned	YES		NULL	
Individual_Product_Price	decimal(5,2) unsigned	NO		NULL	
Expiry_Date	date	YES		NULL	

```
5 rows in set (0.01 sec)
```

The above picture displays the structure of the **Products** table using the DESCRIBE Products; command in MySQL. It shows the fields, data types, constraints, and attributes associated with each column. The **Product_ID** field is defined as an auto-incrementing primary key, ensuring a unique identification number for every product. The **Product_Name** field stores the name of the product, while **Total_Quantity_Available** stores the stock quantity. The **Individual_Product_Price** field stores the price of a single unit of the product, and the **Expiry_Date** field stores the product's expiry date. This table structure forms the foundation of the Super Market Management System and enables efficient storage, retrieval, and management of product information.

❖ PROGRAM OR CODE FOR THE SUPER MARKET APPLICATION

```
import mysql.connector

from datetime import date,datetime


while True:

    try:

        db=mysql.connector.connect(

            host="localhost",

            user="root",

            password="",

            database="Super_Market")

        c=db.cursor()

        break

    except:

        print("Unable to to connect database")

        while True:

            try:

                a=input("Do you want to try again('y','n'): ")

                break

            except:

                print("Please enter 'y' or 'n'")

        if a=='n':

            exit()

            break

print("SUPER MARKET APPLICATION\n")
```



```
def expiry(x):
    c.execute("select Expiry_Date from products where Product_ID=%s",(x,))
    d=c.fetchone()
    if d and d[0] is not None:
        if d[0]>date.today():
            print("Not expired")
        elif d[0]<=date.today():
            print("Expired")
    else:
        print("No expiry date")

def add_to_list(d,z,q,p):
    if d[2] == 0:
        print("\nProduct out of stock")
    elif z>d[2]:
        print("\nOnly", d[2], "items available")
        z=d[2]
        p.append([d,z])
        print(f"{'ID':>5} | {'Name':<13} | {'Price':>11} | {'Quantity':>8} |  
{'E.C':<13}")
        print("="*63)
        print(f"{'d[0]':>5} | {'d[1]':<13} | {'d[3]':>11} | {'z':>8} | ",end='')
        expiry(q)
    else:
        p.append([d,z])
        print(f"{'ID':>5} | {'Name':<13} | {'Price':>11} | {'Quantity':>8} |  
{'E.C':<13}")
        print("="*63)
        print(f"{'d[0]':>5} | {'d[1]':<13} | {'d[3]':>11} | {'z':>8} | ",end='')
        expiry(q)
```

```
def billing_system():
    p=[]
    t=0
    while True:
        while True:
            try:
                q=int(input("\nEnter the product ID: "))
                break
            except:
                print("Please enter a proper ID")
                continue

        while True:
            try:
                z=int(input("Enter the quantity: "))
                break
            except:
                print("Please enter a proper quantity\n")
                continue

        c.execute("select * from Products where Product_ID=%s",(q,))
        d=c.fetchone()
        if d is not None:
            if d[0] in (i[0][0] for i in p):
                print("Product already added to the list")
                while True:
                    while True:
                        try:
                            a=input("\nDo you want to increase the quantity of this
product(y,n): ").lower()
```

```
        break
    except:
        print("Please enter 'y' or 'n'")
    if a!='y' and a!='n':
        print("Please enter 'y' or 'n'")
        continue
    else:
        break
    if a=='y':
        for k in p:
            if k[0][0] == d[0]:
                if k[1]+z >d[2]:
                    print("\nOnly", d[2], "items available")
                    k[1]=d[2]
                else:
                    k[1] += z
                break
            else:
                print("Quantity not updated")
        else:
            add_to_list(d,z,q,p)
    else:
        print("\nProduct not available")
    while True:
        while True:
            try:
                a=input("\n\nDo you want to add another product(y,n): ").lower()
                break
            except:
                print("Please enter 'y' or 'n'")
```

```
        if a!='y' and a!='n':
            print("Please enter 'y' or 'n'")
            continue
        else:
            break

    if a=='y':
        continue
    else:
        print("\nFinal bill is:\n")
        print(f"{'ID':>5} | {'Name':<13} | {'Price':>12} | {'Quantity':>8} | {'E.C':<13}")
        print("="*64)
        for i,z in p:
            print(f"{'i[0]:>5} | {'i[1]:<13} | {'i[3]:>12} | {'z:>8} | ",end='')
            expiry(i[0])

            t+=i[3]*z
            current_quantity = i[2] - z

            c.execute("update Products set Total_Quantity_Available=%s where Product_ID=%s", (current_quantity, i[0]))
            db.commit()
            print("\nTotal Price:",t)
            p.clear()
            t=0
            print("\n")
            break

def add_product():
    while True:
        while True:
            try:
```

```
n=input("\nEnter the product name: ").title()
break
except:
    print("Please enter a proper name")
c.execute("select * from Products where Product_Name=%s",(n,))
d=c.fetchone()
if d is not None:
    print("Product already available\n")
    while True:
        while True:
            try:
                f=input("Do you want to increase the product's quantity
instead('y','n'): ")
                break
            except:
                print("Please enter 'y' or 'n'")
                continue
        if f!='y' and f!='n':
            print("Please enter 'y' or 'n'\n")
            continue
        else:
            break
    if f=='y':
        while True:
            try:
                t=int(input("\nEnter the total quantity: "))
                break
            except:
                print("Please enter a proper quantity")
        c.execute("update products set Total_Quantity_Available=%s where
Product_Name=%s",(d[2]+t,n))
```

```
        db.commit()

        print("\nProduct's Quantity has be successfully Increased\n")
    else:
        print("\n")
else:
    while True:
        try:
            t=int(input("Enter the total quantity: "))

            break
        except:
            print("Please enter a proper quantity\n")
    while True:
        try:
            p=float(input("Enter the individual product price: "))

            break
        except:
            print("Please enter a proper price value\n")
    while True:
        try:
            e=input("Enter the expiry date of the product(YYYY-MM-DD): ")

            if e=="":
                e=None
                break

            e=datetime.strptime(e,"%Y-%m-%d").date()

            break
        except:
            print("Invalid date format\n")

            continue

    c.execute("insert into Products(Product_Name, Total_Quantity_Available,
Individual_Product_Price, Expiry_Date) values(%s,%s,%s,%s)",(n,t,p,e))
```

```
        db.commit()

        print("\nProduct",n,"successfully added to the database of the Super
Market\n\n")

    while True:
        while True:
            try:
                a=input("Do you want to add another product(y,n): ").lower()
                break
            except:
                print("Please enter 'y' or 'n'")
        if a!='y' and a!='n':
            print("Please enter 'y' or 'n'\n")
            continue
        else:
            break
    if a=='y':
        continue
    else:
        print("\n")
        break

def remove_product():
    while True:
        while True:
            try:
                r=input("Enter the product name or ID to be removed: ").title()
                break
            except:
                print("Please enter a proper ID or name")
        if r.isdigit():
```

```
c.execute("select * from Products where Product_ID=%s",(r,))
d=c.fetchone()
if d:
    c.execute("delete from Products where Product_ID=%s",(r,))
    print("\nProduct with ID",r,"is successfully removed from the
database of the Super Market\n\n")
else:
    print("Product not available\n")
else:
    c.execute("select * from Products where Product_Name=%s",(r,))
    d=c.fetchone()
    if d:
        c.execute("delete from Products where Product_Name=%s",(r,))
        print("\nProduct with name",r,"is successfully removed from the
database of the Super Market\n\n")
    else:
        print("Product not available\n")
db.commit()
while True:
    while True:
        try:
            a=input("Do you want to remove another product(y,n): ").lower()
            break
        except:
            print("Please enter 'y' or 'n'")
    if a!='y' and a!='n':
        print("Please enter 'y' or 'n'\n")
        continue
    else:
        break
    if a=='y':
```



```
        continue
    else:
        break

def search_product():
    while True:
        try:
            s=input("\nEnter the product name or ID to search for it: ").title()
            break
        except:
            print("Please enter a proper ID or name")
    if s.isdigit():
        c.execute("select * from Products where Product_ID=%s",(s,))
    else:
        c.execute("select * from Products where Product_Name=%s",(s,))
    d=c.fetchone()
    if d is not None:
        print("\n")
        print(f"{'ID':>5} | {'Name':<13} | {'Total Quantity':<14} | {'Price':<11} | {'Expiry Date':<13}")
        print("="*66)
        if d[4] is not None:
            expiry_date =d[4]
        else:
            expiry_date ="N/A"
        print(f"{d[0]:>5} | {d[1]:<13} | {d[2]:<14} | {d[3]:<11} | {expiry_date:<13}")
        print("\n")
    else:
        print("Product not available\n")
```

```
def expiry_checker():
    c.execute("select * from products where Expiry_Date<=curdate()")
    d=c.fetchall()
    if d:
        print("\n")
        print(f"{'ID':>5} | {'Name':<13} | {'Total Quantity':>14} | {'Price':>11} | {'Expiry Date':<13} | {'E.C':<15}")
        print("="*78)
        for i in d:
            expiry_date=str(i[4])
            print(f"{'i[0]:>5} | {'i[1]:<13} | {'i[2]:>14} | {'i[3]:>11} | {expiry_date:<13} | {'Expired':<15}")
            print("\n")
        else:
            print("\nNo expired products\n")

def view_products():
    c.execute("select * from products")
    d=c.fetchall()
    print("\n")
    print(f"{'ID':>5} | {'Name':<13} | {'Total Quantity':>14} | {'Price':>11} | {'Expiry Date':<13}")
    print("="*66)
    if d:
        for i in d:
            i
            if i[4] is not None:
                expiry_date =str(i[4])
            else:
                expiry_date ="N/A"
```

```
        print(f"{i[0]:>5} | {i[1]:<13} | {i[2]:>14} | {i[3]:>11} |  
{expiry_date:<13}")  
        print("\n")  
    else:  
        print("\nNo products\n")  
  
def product_data_update():  
    while True:  
        while True:  
            try:  
                v=input("\nEnter the product ID or name to be updated in the database  
of the Super Market: ").title()  
                break  
            except:  
                print("Please enter a proper ID or name")  
        if v.isdigit():  
            c.execute("select * from Products where Product_ID=%s",(v,))  
        else:  
            c.execute("select * from Products where Product_Name=%s",(v,))  
        d=c.fetchone()  
        if d is not None:  
            print("\n1= Total quantity available")  
            print("2= Individual product price")  
            print("3= Expiry date\n")  
            while True:  
                try:  
                    p=int(input("Enter the integer of the data need to be updated for  
the product in the database of the Super Market: "))  
                    break  
                except:  
                    print("Please enter the proper integer\n")
```

```
        continue

    if p==1:
        while True:
            try:
                q=int(input("\nEnter the quantity to be be updated: "))
                break
            except:
                print("Please enter a proper quantity")

        c.execute("update products set Total_Quantity_Available=%s where
Product_ID=%s",(q,d[0]))

        print("\nQuantity of the product",d[1],"has been successfully
updated")

    elif p==2:
        while True:
            try:
                q=float(input("\nEnter the price to be be updated: "))
                break
            except:
                print("Please enter a proper price value")

        c.execute("update products set Individual_Product_Price=%s where
Product_ID=%s",(q,d[0]))

        print("\nIndividual product price of the product",d[1],"has been
successfully updated")

    elif p==3:
        while True:
            try:
                q=input("\nEnter the expiry date of the product(YYYY-MM-DD):
")

                if q=="":
                    q=None
                    break

                q=datetime.strptime(q,"%Y-%m-%d").date()
```

```
        break
    except:
        print("Invalid date format\n")
        continue

    c.execute("update products set Expiry_Date=%s where
Product_ID=%s",(q,d[0]))

    print("\nExpiry date of the product",d[1],"has been successfully
updated")

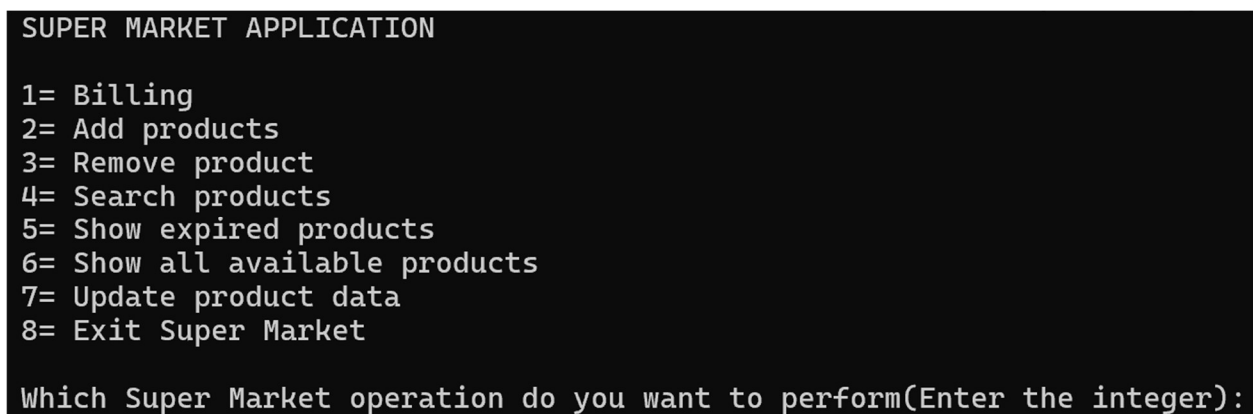
    db.commit()
else:
    print("Product not available\n")
while True:
    while True:
        try:
            a=input("\nDo you want to update another product(y,n): ").lower()
            break
        except:
            print("Please enter 'y' or 'n'\n")
    if a!='y' and a!='n':
        print("Please enter 'y' or 'n'\n")
        continue
    else:
        break
if a=='y':
    continue
else:
    print("\n")
    break

k=range(1,9)
```

```
while True:
    print("1= Billing")
    print("2= Add products")
    print("3= Remove product")
    print("4= Search products")
    print("5= Show expired products")
    print("6= Show all available products")
    print("7= Update product data")
    print("8= Exit Super Market\n")
    try:
        l=int(input("Which Super Market operation do you want to perform(Enter the integer): "))
    except:
        print("Please enter the proper integer\n")
        continue
    if l not in k:
        print("Please enter the integer from the given data\n")
        continue
    if l==1:
        billing_system()
    elif l==2:
        add_product()
    elif l==3:
        remove_product()
    elif l==4:
        search_product()
    elif l==5:
        expiry_checker()
    elif l==6:
        view_products()
```

```
elif l==7:
    product_data_update()
elif l==8:
    print("\nThank you for using the Super Market Application")
    break
c.close()
db.close()
```

❖ OUTPUT

A screenshot of a Python IDLE window showing the output of a program. The text is as follows:

```
SUPER MARKET APPLICATION
1= Billing
2= Add products
3= Remove product
4= Search products
5= Show expired products
6= Show all available products
7= Update product data
8= Exit Super Market
Which Super Market operation do you want to perform(Enter the integer):
```

This picture shows the main menu of the Super Market Application as displayed in Python IDLE. It presents the various operations available to the user, such as billing, adding and removing products, searching products, viewing stock details, checking expired products, updating product data, and exiting the application.

Which Super Market operation do you want to perform(Enter the integer): 6

ID	Name	Total Quantity	Price	Expiry Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	N/A
6	Plates	40	70.00	N/A
7	Toot Brush	60	40.00	N/A
8	Toot Paste	60	10.00	N/A
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	N/A
12	Glue	5	10.00	N/A
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	N/A

This picture shows the “Show All Available Products” feature of the Super Market Application as displayed in Python IDLE. It presents a complete list of products stored in the database along with their Product ID, Product Name, Available Quantity, Individual Price, and Expiry Date, allowing the user to view and monitor the current inventory of the supermarket.

Which Super Market operation do you want to perform(Enter the integer): 5

ID	Name	Total Quantity	Price	Expiry Date	E.C
2	Milk	20	45.50	2026-05-08	Expired
3	Bread	15	30.00	2026-05-08	Expired

This picture shows the “Show Expired Products” feature of the Super Market Application as displayed in Python IDLE. It lists all products whose expiry dates have passed, along with their Product ID, Product Name, Available Quantity, Price, Expiry Date, and Expiry Condition, helping the user identify products that are no longer safe for sale.


```
Which Super Market operation do you want to perform(Enter the integer): 4
Enter the product name or ID to search for it: 5

  ID | Name          | Total Quantity | Price      | Expiry Date
=====
  5 | Bottle         | 30             | 150.00     | N/A
```

```
Which Super Market operation do you want to perform(Enter the integer): 4
Enter the product name or ID to search for it: Bottle

  ID | Name          | Total Quantity | Price      | Expiry Date
=====
  5 | Bottle         | 30             | 150.00     | N/A
```

This picture shows the “Search Product” feature of the Super Market Application as displayed in Python IDLE. It displays the details of a specific product searched by the user, including its Product ID, Product Name, Available Quantity, Price, and Expiry Date, allowing quick retrieval of product information from the database.

```
Which Super Market operation do you want to perform(Enter the integer): 2
Enter the product name: Soap
Enter the total quantity: 30
Enter the individual product price: 60
Enter the expiry date of the product(YYYY-MM-DD): 2027-06-01

Product Soap successfully added to the database of the Super Market

Do you want to add another product(y,n): y
Enter the product name: Soap
Product already available

Do you want to increase the product's quantity instead('y','n'): y
Enter the total quantity: 5

Product's Quantity has be successfully Increased
```

```
mysql> select * from Products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	NULL
12	Glue	5	10.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL
15	Soap	35	60.00	2027-06-01

```
15 rows in set (0.00 sec)
```

Database after adding the product "Soap" and after increasing its quantity

The output shows the product addition feature of the Super Market Management System. A new product named "Soap" is added successfully to the database. When the same product is entered again, the system detects the duplicate entry and offers to increase its quantity. The stock quantity is then updated successfully.

```
Which Super Market operation do you want to perform(Enter the integer): 3
Enter the product name or ID to be removed: 15
```

Product with ID 15 is successfully removed from the database of the Super Market

```
mysql> select * from Products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	NULL
12	Glue	5	10.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

```
14 rows in set (0.01 sec)
```

The output demonstrates the product removal feature of the Super Market Management System. The user selects the delete operation and enters Product ID 15. The program searches the database, removes the corresponding product record, and displays a confirmation message. This ensures accurate inventory management and efficient database maintenance.

```
Which Super Market operation do you want to perform(Enter the integer): 7

Enter the product ID or name to be updated in the database of the Super Market: 11

1= Total quantity available
2= Individual product price
3= Expiry date

Enter the integer of the data need to be updated for the product in the database of the Super Market: 1

Enter the quantity to be be updated: 15

Quantity of the product Pencil Set has been successfully updated

Do you want to update another product(y,n): y

Enter the product ID or name to be updated in the database of the Super Market: 12

1= Total quantity available
2= Individual product price
3= Expiry date

Enter the integer of the data need to be updated for the product in the database of the Super Market: 2

Enter the price to be be updated: 20

Individual product price of the product Glue has been successfully updated

Do you want to update another product(y,n): y

Enter the product ID or name to be updated in the database of the Super Market: 3

1= Total quantity available
2= Individual product price
3= Expiry date

Enter the integer of the data need to be updated for the product in the database of the Super Market: 3

Enter the expiry date of the product(YYYY-MM-DD): 2026-06-03

Expiry date of the product Bread has been successfully updated
```

```
mysql> select * from products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	NULL
12	Glue	5	10.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

14 rows in set (0.00 sec)

```
mysql> select * from products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-05-08
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	15	20.00	NULL
12	Glue	5	20.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

14 rows in set (0.00 sec)

Before and After Updating the Products

The output demonstrates the product update feature of the Super Market Management System. The user updates different product details, including quantity, price, and expiry date, by selecting the appropriate option. The program modifies the corresponding records in the database and confirms each successful update, ensuring accurate and up-to-date inventory information.

```
Which Super Market operation do you want to perform(Enter the integer): 1
Enter the product ID: 11
Enter the quantity: 10
  ID | Name      | Price | Quantity | E.C
=====
  11 | Pencil Set | 20.00 |      10 | No expiry date

Do you want to add another product(y,n): y

Enter the product ID: 11
Enter the quantity: 3
Product already added to the list

Do you want to increase the quantity of this product(y,n): y

Do you want to add another product(y,n): y

Enter the product ID: 11
Enter the quantity: 7
Product already added to the list

Do you want to increase the quantity of this product(y,n): y

Only 15 items available

Do you want to add another product(y,n): y

Enter the product ID: 3
Enter the quantity: 5
  ID | Name      | Price | Quantity | E.C
=====
   3 | Bread     | 30.00 |       5 | Not expired

Do you want to add another product(y,n): n

Final bill is:

  ID | Name      | Price | Quantity | E.C
=====
  11 | Pencil Set | 20.00 |      15 | No expiry date
   3 | Bread     | 30.00 |       5 | Not expired

Total Price: 450.00
```

```
mysql> select * from products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	15	30.00	2026-06-03
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	15	20.00	NULL
12	Glue	5	20.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

14 rows in set (0.00 sec)

```
mysql> select * from products;
```

Product_ID	Product_Name	Total_Quantity_Available	Individual_Product_Price	Expiry_Date
1	Chocolate	70	50.00	2027-05-09
2	Milk	20	45.50	2026-05-08
3	Bread	10	30.00	2026-06-03
4	Biscuits	50	20.00	2026-08-20
5	Bottle	30	150.00	NULL
6	Plates	40	70.00	NULL
7	Toot Brush	60	40.00	NULL
8	Toot Paste	60	10.00	NULL
9	Cake	40	20.00	2026-09-26
10	Juice	20	20.00	2026-10-02
11	Pencil Set	0	20.00	NULL
12	Glue	5	20.00	NULL
13	Butter	100	60.00	2026-09-06
14	Books	40	100.00	NULL

14 rows in set (0.00 sec)

Database Before and After billing the products

The output demonstrates the billing and sales feature of the Super Market Management System. The user selects products and quantities for purchase. The program prevents duplicate entries, allows quantity updates, checks stock availability, and generates a final bill showing purchased items, quantities, prices, expiry status, and the total amount payable.

```
Which Super Market operation do you want to perform(Enter the integer): 8  
Thank you for using the Super Market Application  
C:\Users\Menshafi>
```

The output demonstrates the exit feature of the Super Market Management System. When the user selects option 8, the program terminates the application safely and displays a thank-you message. This provides a proper ending to the session, ensuring that all operations are completed before the user exits the system.

❖ BIBLIOGRAPHY

- Computer Science with Python by Sumita Arora
- <https://www.google.com/>
- <https://github.com/Menshafi-2629/Python-Projects>
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❖ CONCLUSION

The Super Market Management System was successfully developed using Python and MySQL to simplify and automate various supermarket operations. The project effectively manages product information, billing, inventory records, product searches, expiry checking, and product updates through a database-driven approach.

One of the most important features of this project is its ability to display the expiry status of products during billing, helping ensure that customers are informed whether a product is expired, not expired, or does not have an expiry date. This contributes to customer safety and responsible inventory management.

The project also incorporates a trial-and-error approach through extensive input validation and exception handling. As a result, the program does not terminate or break when invalid inputs or unexpected errors occur. Instead, it guides the user to provide correct information and continues execution smoothly, making the application more reliable and user-friendly.

Although the current version operates through a text-based interface in Python IDLE, it can be further enhanced by implementing a Graphical User Interface (GUI) for a more professional user experience.

Overall, this project demonstrates the practical application of Python programming, database management, and error-handling techniques in developing an efficient and dependable supermarket management solution.

